

Evidence-Grounded Local-LLM Explanations for Fraud Alert Review

A fail-closed explanation delivery boundary
built on detector evidence, SHAP reason
codes, deterministic validation, and fallback.

Ng Yi Zhen | 23076003

FR

Fraud Review

GUARDED EXPLANATION SYSTEM

WORK

Alert Queue

50 default, ULB labelled

ASSURANCE

Explanation Assurance

Guardrails and fallback

Evaluation Results

LLM first, detector supporting

LOCAL ANALYST SYSTEM

Evidence verified

Narrative service available

Case workflow ready

ANALYST WORKBENCH

Alert Queue

S0 explanation cases by default; ULB supporting benchmark available

EVIDENCE VERIFIED

Evidence details

66 unreviewed6 in review0 follow-up4 fallback briefs

ANALYST WORK QUEUE

Alert Queue

Review readable S0 alerts through model evidence, a guarded local-LLM brief, and a human routing decision. ULB alerts remain available as supporting real-data detector evidence.

66 awaiting review

Review next alert

UNREVIEWED66IN REVIEW6FOLLOW-UP0CLOSED4

25 primary explanation cases

S0 synthetic readable evidence - default review queue - ranks within the S0 detector.

51 supporting benchmark alerts

ULB real-data detector evidence - anonymous PCA features - not comparable with S0 ranks.

72/76 guarded briefs passed

4 deterministic fallbacks - the local-LLM explanation path remains separate from supporting detector evidence

Model-flagged alerts

76 shown - all labelled evidence sources

Search case, feature or am

Source

All sources

Review state

All states

Explanation

All delivery states

Sort

Work priority

STATE	SOURCE	TRANSACTION / CASE	RANK	MODEL EVIDENCE	BRIEF
In review	RESEARCH	Case 42769 Amount 118.30	#2 of 51	V14 T V12 T V10 T	Verified Jul 16, 12:55 AM Start review
In review	RESEARCH	Case 122479 Amount 219.80	#6 of 51	V14 T V12 T V10 T	Verified Jul 24, 09:38 AM Start review
In review	RESEARCH	Case 43168 Amount 273.01	#7 of 51	V14 T V12 T V10 T	Verified Jul 24, 10:08 AM Start review
In review	RESEARCH	Case 151887 Amount 33.59	#8 of 51	V14 T V12 T V10 T	Verified Jul 24, 10:25 AM Start review
In review	RESEARCH	Case 154234 Amount 1.00	#42 of 51	V14 T V4 T V12 T	Fallback Jul 24, 08:56 AM Start review
In review	RESEARCH	Case 215132 Amount 733.31	#50 of 51	V14 T V14 L V10 T	Fallback Start review

Local analyst workbench using frozen, source-labelled evidence

- Fraud detection is highly imbalanced; accuracy is misleading.
- Explanations can mislead if generated text is treated as evidence.
- Prior work shows the need for faithful narrative explanation, not just readable prose.
- Local LLMs may reduce external exposure, but local generation can still drift from evidence.
- The project asks how to keep generated explanations inside the model-evidence contract.

Rare-event detection

Accuracy hides the fraud minority; AP, recall, precision, and F1 matter.

Technical attribution

SHAP gives signed model evidence, but raw attribution is not analyst prose.

Narrative risk

LLM text may omit, add, or reverse evidence if delivered unchecked.

Controlled delivery

Deterministic guardrails either accept the brief or return fallback evidence.

Visible literature gap

AlMarri (2025): SHAP-faithfulness concern • Zyte (2024): narrative explanations • Rebedea (2023): programmable guardrails

This project measures the delivery boundary between evidence and generated articulation.

- RQ1: How effectively do deterministic guardrails detect local-LLM output violations and enforce fallback?
- RQ2: What does guarded local-LLM delivery add, preserve, or lose compared with SHAP reason codes and deterministic briefs?
- ULB is the real anonymised benchmark and anonymous-feature prompt stress test.
- S0 is the synthetic readable-feature study for semantic and operational evaluation.
- They are two evidence roles in one study, not two competing detector datasets.
- S0 follows Fraud Detection Handbook concepts to make feature semantics inspectable.

ULB benchmark

Real anonymised credit-card data

Role: detector benchmark, SHAP chain, anonymous-feature prompt stress test

Boundary: no readable business semantics

S0 semantic study

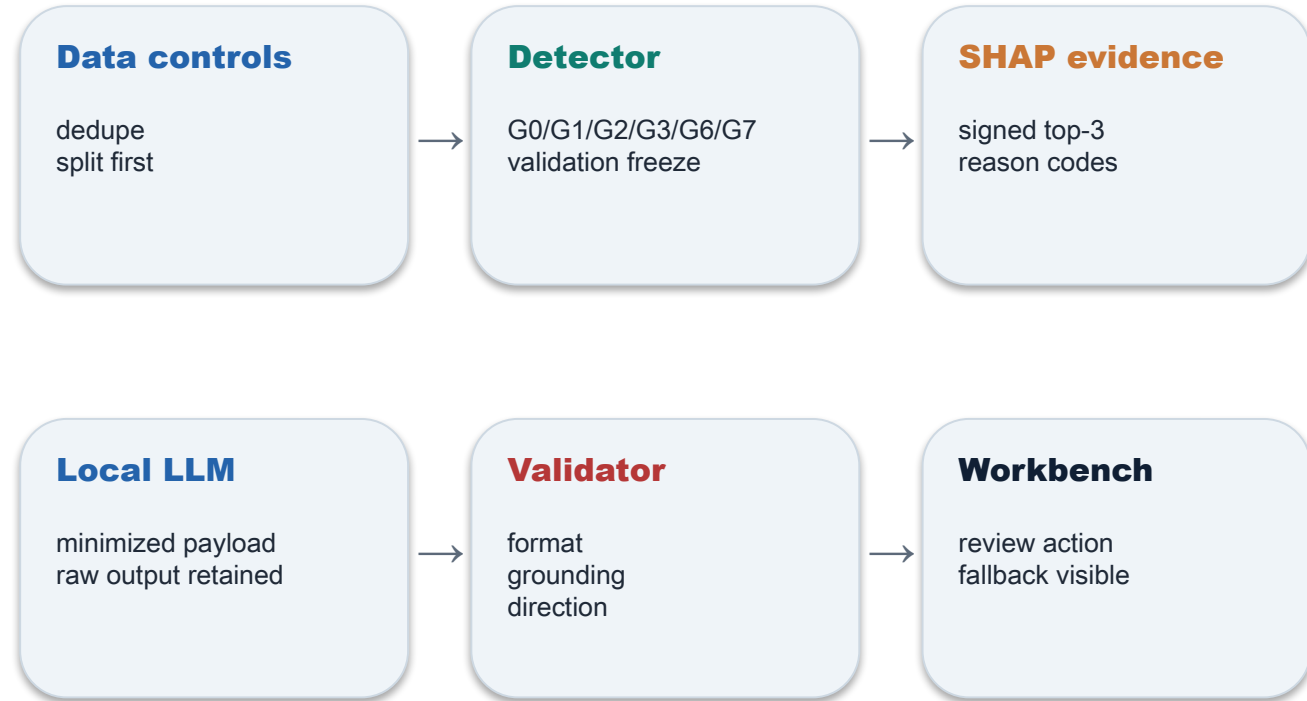
Synthetic readable transaction stream

Role: operational explanation, workbench review, human pilot stimuli

Source basis: Le Borgne et al. (2022)

One contribution: evidence-governed local-LLM explanation delivery

- Deduplicate source data and create stable case IDs.
- Split before scaling, SMOTE, autoencoder fitting, threshold selection, and SHAP.
- Train six detector groups across five fixed seeds.
- Freeze the selected detector and generate signed SHAP reason codes.
- Generate local-LLM briefs, validate them, and fall back when needed.
- Consume the same artifacts in the React/FastAPI workbench.



Core rule: the generated brief can never change the stored evidence.

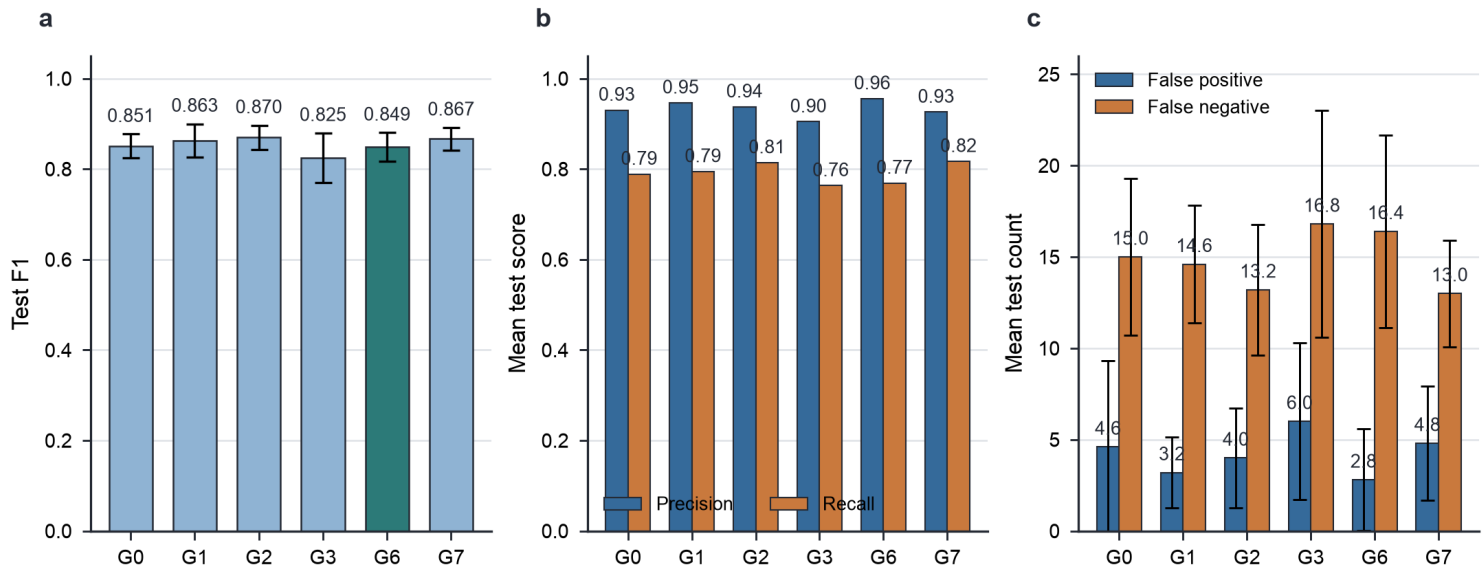
- Groups: G0, G1, G2, G3, G6, and G7 over seeds 42 to 46.
- G0: original features with XGBoost.
- G1/G3: training-only SMOTE.
- G2/G3/G7: autoencoder reconstruction or latent features.
- G6: cost-sensitive XGBoost using `scale_pos_weight`.
- Detector comparison is supporting evidence, not the main novelty.

Group	Input	Imbalance
G0	Original features	None
G1	Original features	SMOTE
G2	Original + AE reconstruction	None
G3	Original + AE reconstruction	SMOTE
G6	Original features	Cost-sensitive
G7	Original + AE latent features	None

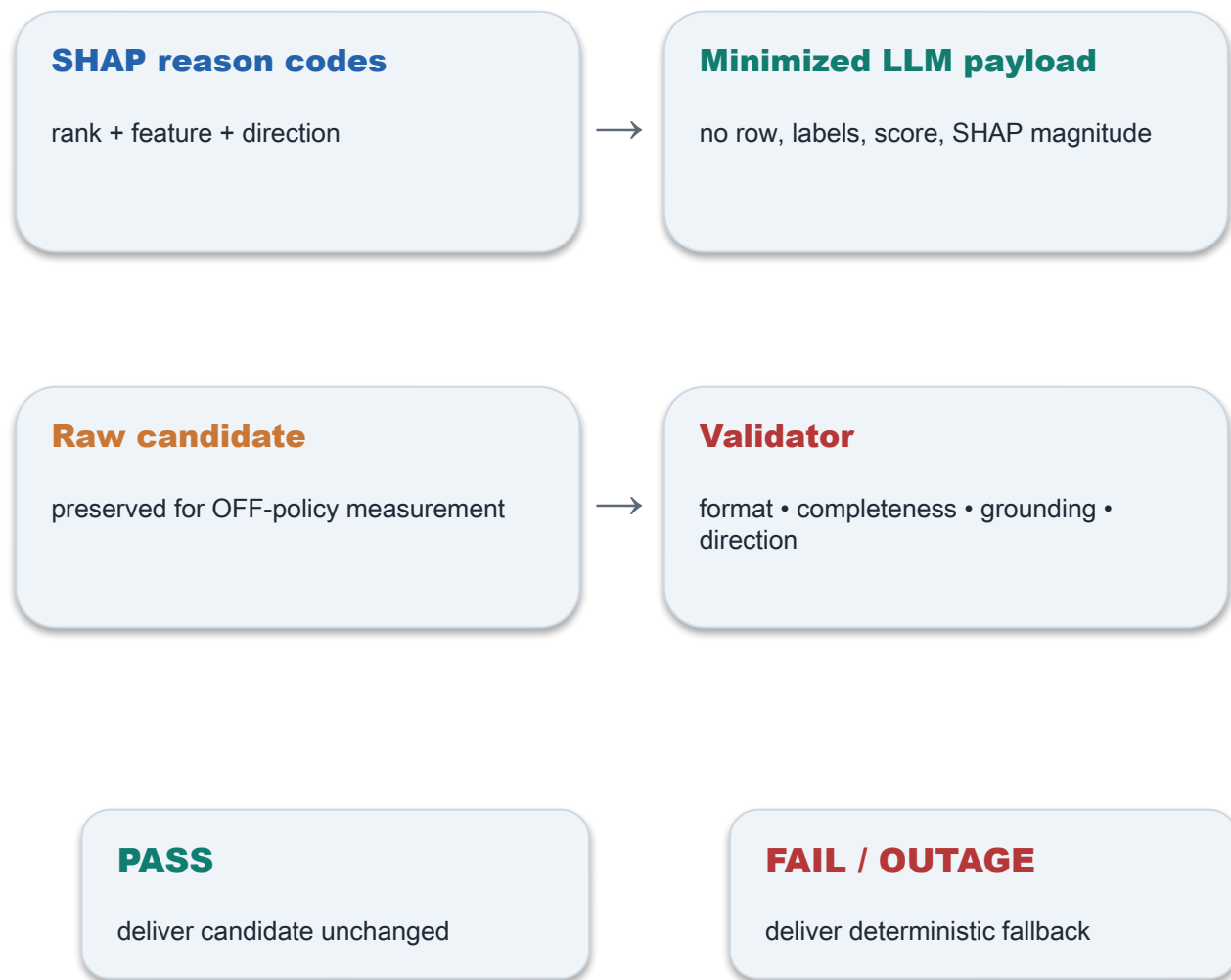
Selection discipline

Detector comparison supplies a reproducible evidence source; it is not the project novelty.

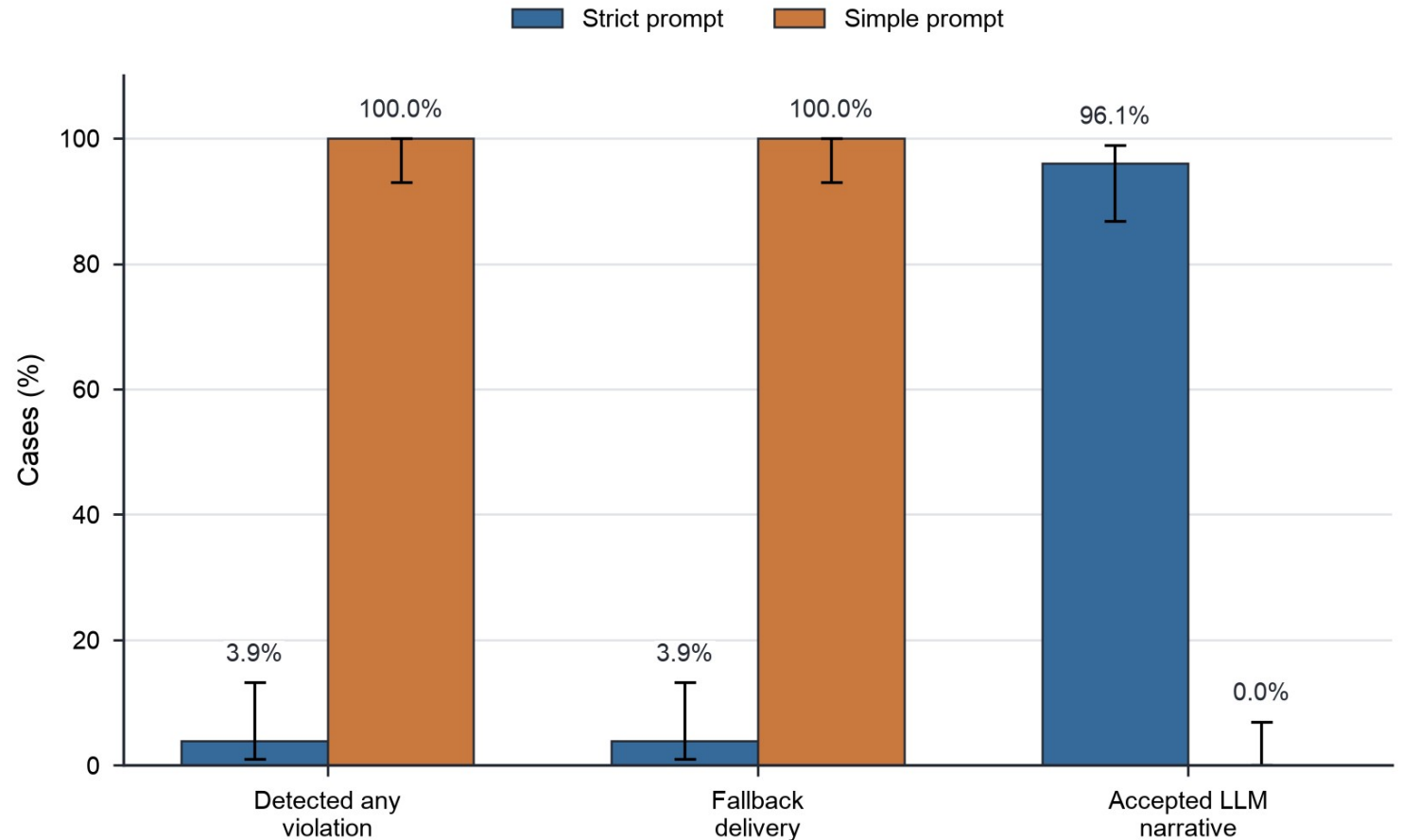
- G6 had the numerically highest mean test AP: 0.855214 ± 0.027097 .
- G2 had the highest mean F1: 0.870054 ± 0.026465 .
- G7 had the highest mean recall: 0.816901 ± 0.041063 .
- G6 had the highest mean precision and lowest mean false-positive count.
- No group led every metric, and no ranked-winner claim is made.



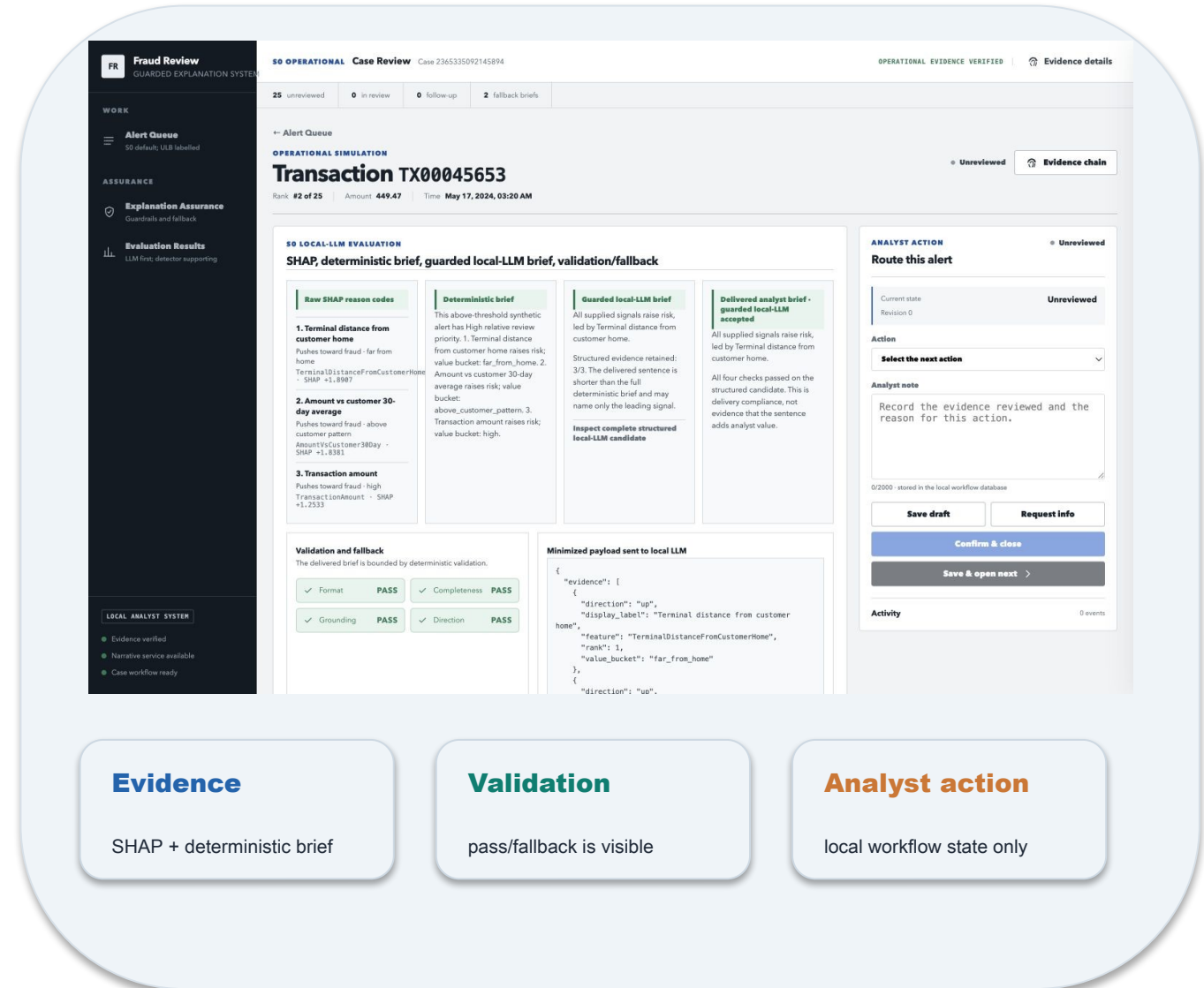
- G4 generated signed SHAP reason codes for 51 flagged ULB test cases.
- The LLM payload excluded raw rows, exact values, labels, probabilities, and SHAP magnitudes.
- OFF-policy analysis measures raw LLM violations.
- ON-policy delivery validates the same raw output and falls back on failure.
- Fallback is generated from reason codes, not from rejected prose.



- ULB strict prompt: 2/51 detected-any violations, 3.92% [1.08%, 13.22%].
- ULB simple prompt: 51/51 detected-any violations, 100% [93.00%, 100%].
- Every detected failure activated deterministic fallback.
- Corpus calibration intercepted 330/330 ULB attacks and accepted 318/318 faithful controls.
- Blinded manual audit: 0/49 delivered strict narratives had a semantic violation [0%, 7.27%].
- The audit used one student reviewer; it is bounded evidence, not universal proof.



- S0 contains 50,000 synthetic transactions with readable operational fields.
- S0 test AP was 0.544017; precision 0.720000, recall 0.400000, F1 0.514286.
- The selected explanation set contained 25 above-threshold alerts.
- Guarded local-LLM briefs passed validation for 23/25 and fell back for 2/25.
- The application presents one source-labelled queue and separates Work from Model Evidence.



Fraud Review
GUARDED EXPLANATION SYSTEM

WORK
Alert Queue
50 default; USB labelled

ASSURANCE
Explanation Assurance
Guardrails and fallback
Evaluation Results
LLM first; detector supporting

LOCAL ANALYST SYSTEM
Evidence verified
Narrative service available
Case workflow ready

S0 OPERATIONAL Case Review Case 236535092145894
25 unreviewed | 0 in review | 0 follow-up | 2 fallback briefs

Alert Queue
OPERATIONAL SIMULATION
Transaction TX00045653
Rank #2 of 25 | Amount 449.47 | Time May 17, 2024, 03:20 AM

S0 LOCAL-LLM EVALUATION
SHAP, deterministic brief, guarded local-LLM brief, validation/fallback

Raw SHAP reason codes
1. Terminal distance from customer home
Pushes toward fraud - far from home
TerminalDistanceFromCustomerHome - SHAP +1.8987
2. Amount vs customer 30-day average
Pushes toward fraud - above customer pattern
AmountVsCustomer30Day - SHAP +1.8381
3. Transaction amount
Pushes toward fraud - high TransactionAmount - SHAP +1.2533

Deterministic brief
This above-threshold synthetic alert has High relative review priority. 1. Terminal distance from customer home raises risk; value bucket: far_from_home. 2. Amount vs customer 30-day average raises risk; value bucket: above_customer_pattern. 3. Transaction amount raises risk; value bucket: high.

Guarded local-LLM brief
All supplied signals raise risk, led by Terminal distance from customer home.
Structured evidence retained: 3/3. The delivered sentence is shorter than the full deterministic brief and may name only the leading signal.
Inspect complete structured local-LLM candidate

Delivered analyst brief - guarded local-LLM accepted
All supplied signals raise risk, led by Terminal distance from customer home.
All four checks passed on the structured candidate. This is delivery compliance, not evidence that the sentence adds analyst value.

Validation and fallback
The delivered brief is bounded by deterministic validation.

✓ Format	PASS	✓ Completeness	PASS
✓ Grounding	PASS	✓ Direction	PASS

Minimized payload sent to local LLM

```
{
  "evidence": [
    {
      "direction": "up",
      "display_label": "Terminal distance from customer home",
      "feature": "TerminalDistanceFromCustomerHome",
      "rank": 1,
      "value_bucket": "far_from_home"
    },
    {
      "direction": "up",

```

Analyst Action
Route this alert
Current state: Revision 0
Unreviewed
Action: Select the next action
Analyst note: Record the evidence reviewed and the reason for this action.
0/2000 - stored in the local workflow database
Save draft | Request info
Confirm & close
Save & open next >
Activity: 0 events

Evidence
SHAP + deterministic brief

Validation
pass/fallback is visible

Analyst action
local workflow state only

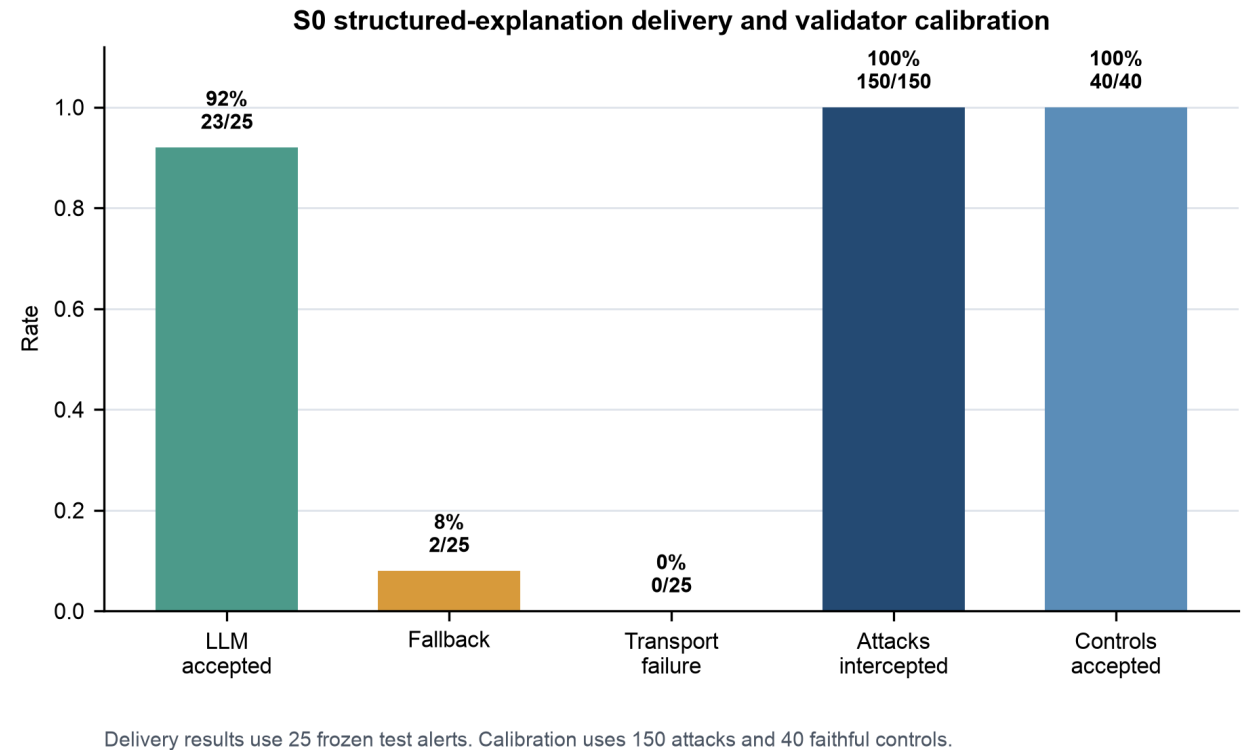
- The deterministic S0 brief preserved all three evidence items.
- All 23 accepted guarded-LLM summaries selected the shorter permitted option.
- Accepted guarded summaries averaged 12 words and one named evidence item.
- Deterministic briefs averaged 39.09 words and three named evidence items.
- Participants more often selected the LLM brief as clearest, but it did not add evidence detail.

39.09

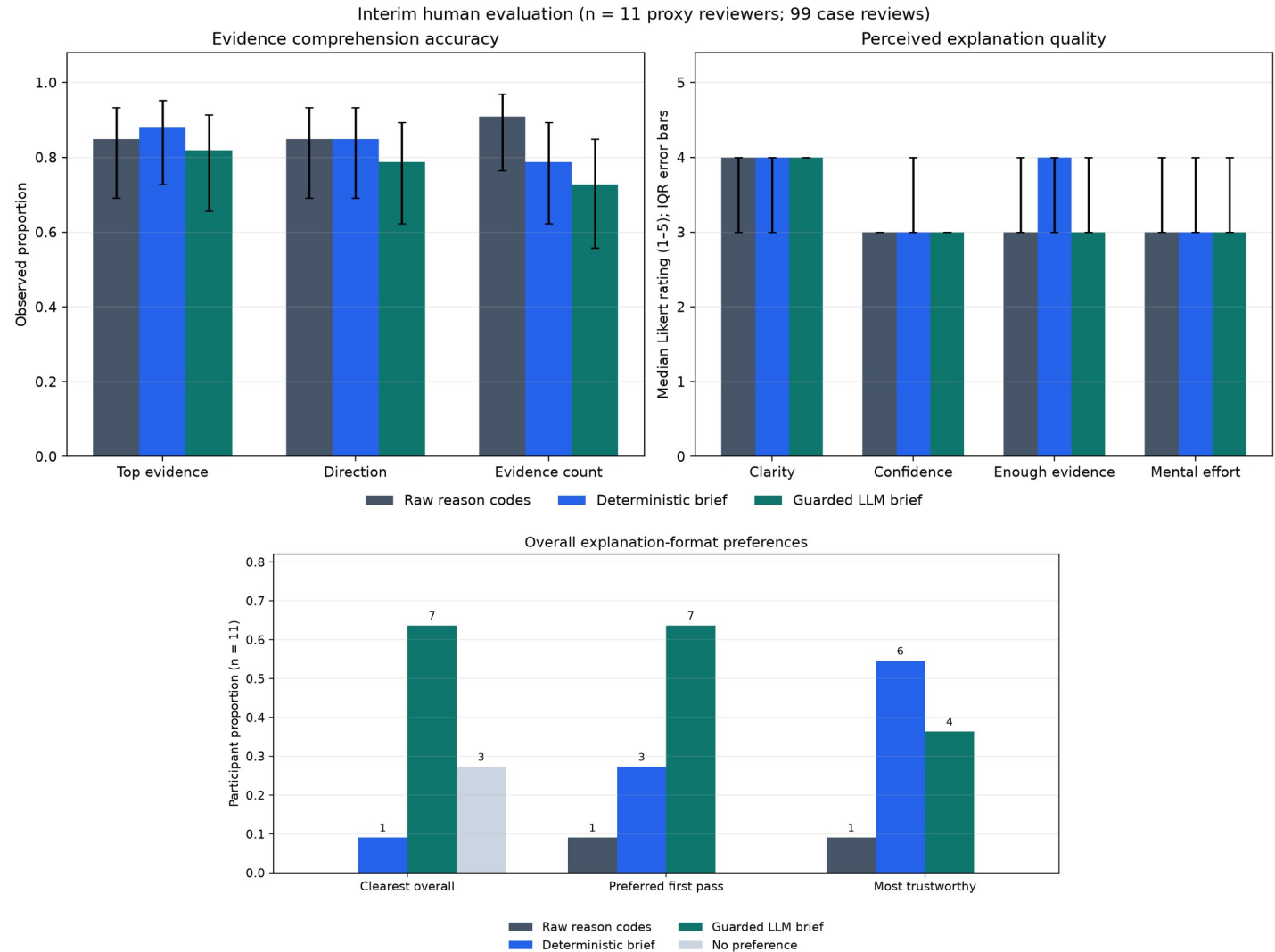
words | 3 evidence items

12.00

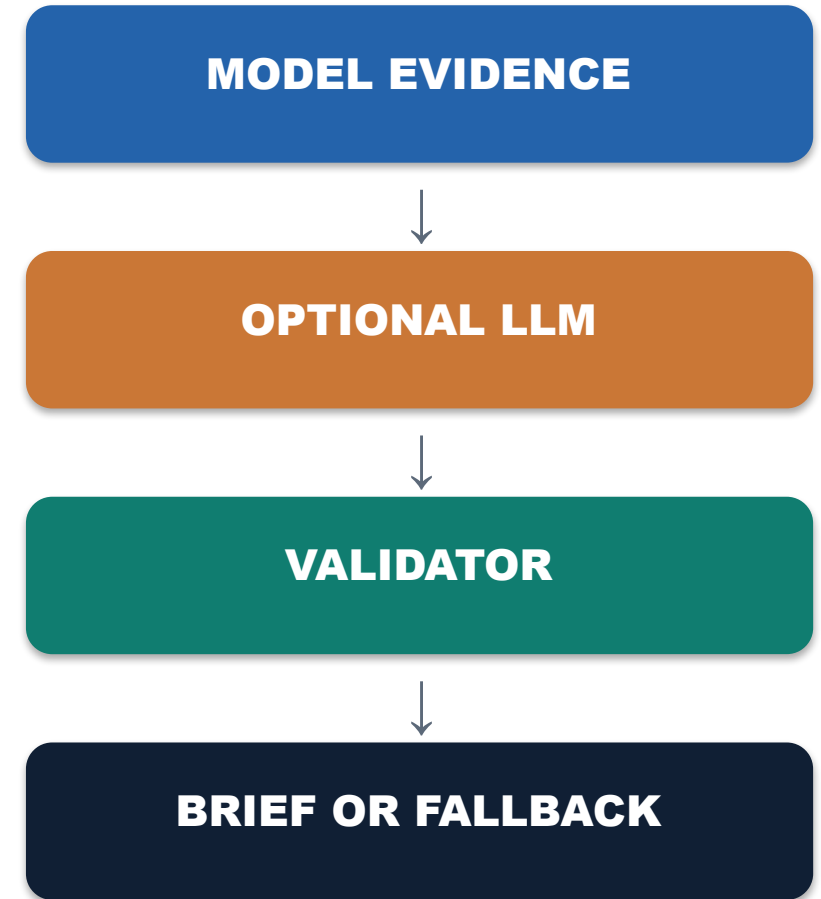
words | 1 evidence item



- Eleven proxy reviewers completed 99 case reviews.
- Accuracy did not favour the guarded LLM brief on any comprehension check.
- Median clarity was 4/5 for all three formats.
- Guarded LLM brief was selected as clearest and preferred first-pass format by 7/11.
- Deterministic brief was selected as most trustworthy by 6/11.
- Interpretation: readability preference did not equal stronger evidence understanding.



- Main contribution: a provenance-linked fail-closed delivery boundary for local-LLM fraud-alert briefs.
- Detector result: autoencoder features did not show a clear advantage over baseline.
- Explanation result: guardrails can measure and block detected violations under tested contracts.
- Human result: perceived clarity and trust do not fully align.
- Societal/environmental relevance: local, minimized, fallback-first design may reduce unsupported explanation exposure and unnecessary model calls.
- Next work: independent audit replication, professional analyst study, additional local models, and real interpretable data.



Readable text never replaces the evidence layer.